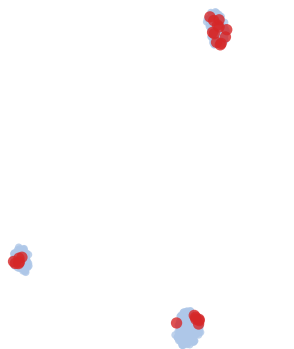


circMAPK1 × miR-12119 | UMAP | Pretrained RNA language models | colored by binding annotation

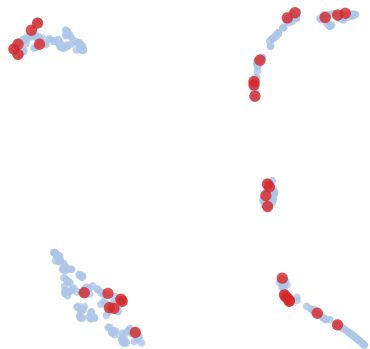
circRNA only

**RNABERT
(fine-tuned)**



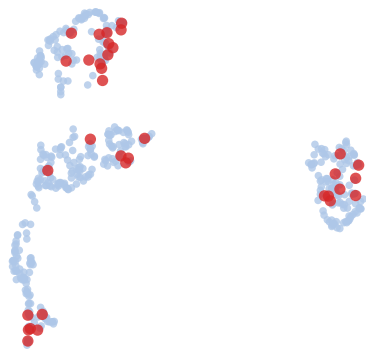
Sep=2.375 Sil=0.017

**RNAErnie
(fine-tuned)**



Sep=1.202 Sil=-0.015

**RNAMSM
(fine-tuned)**



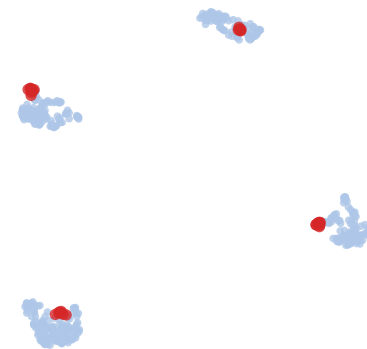
Sep=0.983 Sil=0.033

**RNA-FM
(fine-tuned)**



Sep=0.988 Sil=0.026

**circMAC
(ours)**

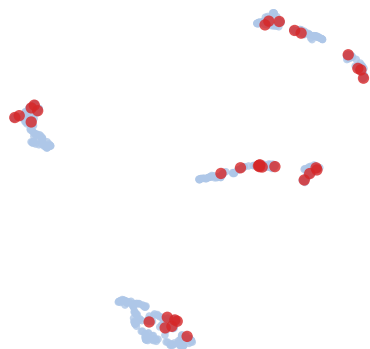


Sep=2.128 Sil=-0.014

circRNA + miRNA



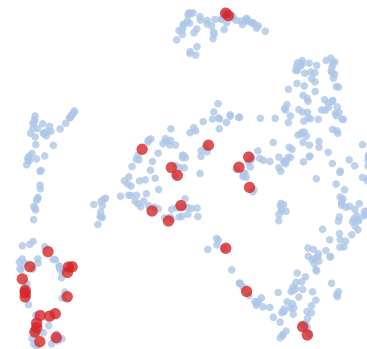
Sep=2.698 Sil=0.038



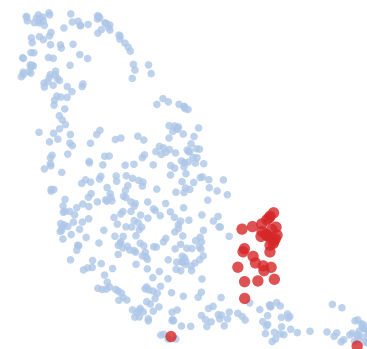
Sep=2.399 Sil=-0.015



Sep=1.977 Sil=-0.023



Sep=3.327 Sil=0.107



Sep=2.844 Sil=0.093

● Binding ● Non-binding